

WASP-135b

R-0204

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_230914 · created 2026-07-16T22:48:21+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460201.6672 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.011
Transit depth (Rp/Rs)^2	3.08 +/- 0.39 [%]
Orbital Inclination (inc)	82.14 +/- 1.5
Airmass coefficient 1 (a1)	1.03 +/- 0.02
Airmass coefficient 2 (a2)	-0.036 +/- 0.043
Scatter in the residuals of the lightcurve fit is	1.93 %
Best Comparison Star	#2 - [363, 212]
Optimal Aperture	3.54
Optimal Annulus	5.5
Transit Duration (day)	0.071 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.175 $\pm$ 0.011 vs 0.139 (deviation 3.27 σ)
Duration fit vs published	0.071 d vs 0.0687 d (deviation 0.2 σ)
Mid-time O–C	-19.3 min
Depth SNR	15.9
Residual scatter	1.93 %

Light curve

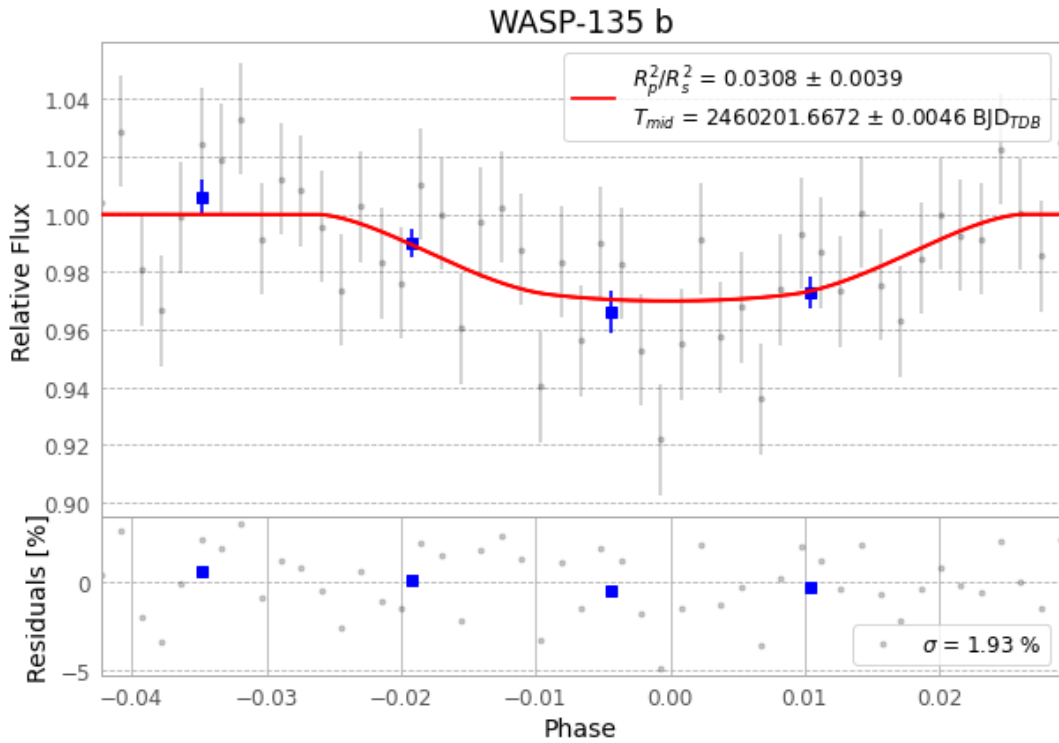


Figure 1 — FinalLightCurve_WASP-135 b_2023-09-13.png (SHA-256 484579a128cb7bed...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [272, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b0c098a42a4498d7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

49 FITS frames (32 MB), WASP-135230914023315.FITS → WASP-135230914045715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-230914.log (8d97aa487f1f...), FinalLightCurve_WASP-135 b_2023-09-13.png (484579a128cb...), FinalLightCurve_WASP-135 b_2023-09-13.csv (ca273df3b403...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 22:48Z.